Final Report

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Introduction

Web mapping platforms like Google Maps and Apple Maps help users find the most efficient routes to their destinations through various travel modes. However, these platforms often overlook the unique characteristics of certain cities. For instance, Minneapolis boasts the most extensive skyway network in the country, spanning over 10 miles and connecting more than 80 blocks and 150 buildings. This network links key locations such as hotels, restaurants, shops, and parking areas. Given the exceptionally cold winters, the climate-controlled skyways are particularly beneficial.

Currently, these web mapping platforms only show walking routes along sidewalks and do not consider the entire network of walkways above ground. The primary goal of this project is to develop a website that generates routes based on user-selected origin and destination points of interest (POIs) with several options, including:

- The shortest path
- The path with the most nearby retail establishments
- The path with the most nearby food establishments
- The path with the most nearby open spaces

This website will give both visitors and residents of Minneapolis a valuable option for navigating to their destinations while staying warm and indoors. Additionally, it will allow users to discover multiple stops at retail, food, and open space POIs that are connected to or near the skyway system.

Methodology

The analysis begins by obtaining the pedestrian network using OSMnx, which retrieves OpenStreetMap data and converts it into a NetworkX MultiDiGraph. The use of graph_from_place and the network_type parameter ensures that only walkable edges are retained. The network is reprojected into a local UTM coordinate reference system to allow metric distance calculations. Accurate distance estimation requires consistent CRS alignment, since shortest path solvers rely on numeric edge lengths stored in meters. The Skyway system is also extracted and used to generate a bounding buffer polygon. This buffer acts as a spatial mask, reducing the graph to a smaller induced subgraph while preserving connectivity properties. The truncation uses polygon-based edge filtering via truncate_graph_polygon,

which removes nodes and edges outside the buffer while retaining the correct topological structure inside.

Parks, restaurants, and retail amenities are retrieved through feature-based queries. One major caveat to mention is that only features with proper names were incorporated into this project. This left us with a total of 79 origin/destination nodes: 6 retail, 63 food, and 10 open space locations. Because amenities may be polygons, multipolygons, or points, the dataset is geometrically standardized by converting all geometries to representative points. This avoids spatial ambiguity and simplifies nearest-edge assignment. Spatial indexing is handled internally by OSMnx using Shapely's STRtree, which accelerates nearest-edge computation by reducing point-to-segment distance checks. The spatial clip ensures that only amenities within the Skyway buffer influence routing.

Each amenity point must be mapped to its nearest edge. For an amenity point and a candidate edge represented as a line segment, the closest point is computed via ox.distance.nearest_edges().

The edge with the global minimal distance receives the amenity assignment. Edge attributes parks_count, restaurants_count, and retail_count store these frequencies. Because the routing graph is a MultiDiGraph, each directional edge is counted separately. This produces an enriched attribute tensor $A(u, v, k)$ that stores:

$$A(u, v, k) = [l_x, n_o, n_f, n_r] \; where$$
$$l_x = length\ of\ edge\ x$$
$$n_o = count\ of\ open\ spaces\ within\ 0.1\ miles\ of\ edge\ x$$
$$n_f = count\ of\ restaurants\ within\ 0.1\ miles\ of\ edge\ x$$
$$n_r = count\ of\ retail\ within\ 0.1\ miles\ of\ edge\ x$$

Routing incorporates a modified function that adjusts each edge's cost C according to amenity presence. The base cost is geometric distance:

$$C_x = l_x \; where$$

$l_x = length\ of\ edge\ x$

For amenity-aware routing, the cost C for selected amenity A becomes:

$$C_{x,n} = \frac{l_x}{1 + \alpha * n_A} \; where$$
$$l_x = length\ of\ edge\ x$$
$$\alpha = amenity\ influence\ coefficient\ (equal\ to\ 4)$$

$$n_A = count\ of\ selected\ amenity\ A\ within\ 0.1\ miles\ of\ edge\ x$$

The form of the function is based on diminishing marginal cost: the presence of additional amenities reduces edge weight, but the magnitude of the reduction decreases as c increases. This acts analogously to a concave utility function. If $n_A = 0$, a penalty multiplier is used:

$$C_{x,n} = 2.5 * l_x\ where$$
$$l_x = length\ of\ edge\ x$$

This creates a cost surface in which amenity-rich paths behave as low-cost corridors, steering the shortest-path solver toward them even when they deviate from the geometric minimum-distance path. This formulation constitutes a multi-objective optimization that approximates a weighted sum of distance and environmental quality.

The routing problem is solved using Dijkstra's algorithm, where appropriate. Because OSMnx stores edge lengths in meters and the modified cost remains positive, Dijkstra remains valid. The weighted graph $G' = (V, E, C)$ is fully connected within the buffer, ensuring that the solver can always compute a valid route. The presence of the concave amenity weight function effectively reshapes the graph's cost surface, allowing edges with higher amenity density to become pseudo-shortcuts. In practice, the solver tends to select paths with large clusters of amenities rather than isolated amenities, since the increase in the denominator scales with n. From a computational perspective, routing runs efficiently because the graph has been spatially reduced and weights are precomputed or evaluated rapidly at runtime.

Results

The processed graph reveals sharp gradients in amenity density. Nicollet Mall contains high concentrations of both restaurants and retail amenities, which substantially reduces effective edge weights in these areas. Park amenities are more distributed and often located near riverfront corridors. The enriched network therefore creates amenity corridors that differ across categories.

When n increases, the cost decreases asymptotically, meaning the solver is highly sensitive to amenity clusters. For example, edges with $n_A = 3$ and $\alpha = 4$ yield:

$$C_{x,n} = \frac{l_x}{1 + 4 * 3} = \frac{l_x}{13}$$

This produces dramatic differences in route geometry when compared to the shortest path. For edges where $n_A = 0$, cost increases by a factor of 2.5, ensuring that truly amenity-poor streets are strongly deprioritized.

These effects illustrate how graph weighting transforms the routing landscape and produces routes designed for environmental richness rather than speed.
OSM-derived data are subject to tagging inconsistencies. Because nearest-edge matching relies on geometric shortest distance, amenities located in building interiors may be assigned to adjacent sidewalks. Buffer selection also influences which amenities matter for routing. CRS transformations introduce minimal but non-zero error due to projection distortions, though UTM minimizes this across the study extent.

Conclusion

This project demonstrates how spatial data science, graph theory, and computational geometry can be combined to develop an amenity-aware routing system grounded in open geospatial data. The hybrid mathematical and geospatial approach enables a more realistic representation of pedestrian experience by embedding environmental qualities directly into routing algorithms. The system provides a methodological foundation for analyzing walkability, micro-scale urban form, and experiential routing using enriched networks. Future work can incorporate additional variables such as safety metrics, pedestrian volume, elevation, or dynamic environmental data. The approach presented here is extensible to other cities and scalable with more complex multi-objective optimization frameworks.